Maternal $PM_{2.5}$ and temperature exposures pre-conception and during pregnancy are associated with increased risk of congenital Chagas disease transmission.

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Summary

Background Particulate air pollution (specifically <2·5µm diameter [$PM_{2\cdot5}$]), temperature, and *Trypanosoma cruzi* (*Tc*) each are deleterious to pregnancy. $PM_{2\cdot5}$ crosses the placenta, weakening the trophoblastic layer through inflammation. Sustained exposure to high heat causes chronic intrauterine inflammation, affecting placental formation. *Tc* parasitemia increases during pregnancy, causing placental degradation through inflammation and resulting in congenital Chagas disease (cCD) in approximately one-in-twenty *Tc(+)* pregnancies. The interaction of $PM_{2\cdot5}$ and temperature during pregnancy with cCD is unstudied.

Methods We analyzed a Bolivian pregnancy cohort (2006-2014) including 581 *Tc(+)* pregnancies with 46 congenitally infected infants. European Space Agency-Copernicus estimates of monthly mean wet-bulb temperature and $PM_{2\cdot5}$ were used to create individual monthly pregnancy exposures for 12-months prior-to-birth. Lagged-Weighted Quantile Sum regression analysis was conducted to determine the time-varying association of these exposures with cCD.

Findings cCD was significantly associated with exposure to higher temperatures and $PM_{2\cdot5}$ pre-conception and high $PM_{2\cdot5}$ in the last three months of pregnancy.

Interpretation Systemic inflammation from $PM_{2\cdot5}$ and high temperatures likely depress the already modulated maternal immune system allowing higher parasitemia during pregnancy. Inflammatory responses to $PM_{2\cdot5}$, temperature, and *Tc* may weaken the placental barrier, allowing the parasite to more easily cross the marginal sinus and the trophoblastic layer of the intervillous space and infect the fetus. In urban areas, vector-bone *Tc* transmission is now uncommon; however, there is still a significant incidence of cCD. Mitigating the effects exposure to high $PM_{2\cdot5}$ and temperatures during pregnancy is an important component to preventing cCD.

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Introduction

Climate change is unevenly impacting populations globally, making regions less habitable for humans by reducing crop yields, intensifying droughts, and increasing wildfires, all of which have caused millions of people to migrate.[1] Additionally, populations are shifting from rural to more densely populated urban areas; this shift, combined with global migration, has the potential to modify infectious disease transmission pathways.[2,3] This is especially true for vector-borne diseases for which climate change impacts the spatial distribution and proximity of both human hosts and vectors, intensifying disease amplification and transmission.[4] One vector-borne Neglected Tropical Disease that may experience important shifts in its transmission dynamics is *Trypanosoma cruzi* (*Tc*), the parasite that causes Chagas disease. Both environmental exposures and *Tc* are known to have deleterious effects on pregnancy. Increased *Tc* parasitemia, stress from exposure to high temperatures and pollution, and the typical pregnancy-associated immune system modulation may work in tandem to facilitate the passage of *Tc* through the placenta, resulting in increased incidence of cCD.

Nearly eight million people are infected with *Tc* globally, and approximately one in three untreated people develop chronic cardiac and/or digestive Chagas disease, resulting in significant morbidity and mortality (Figure 1).[5] *Tc* infection is most commonly acquired when an infected triatomine bug defecates while feeding, and the infectious stage of *Tc* contained in the feces is rubbed into the bite lesion or intact mucous membrane such as conjunctiva.[5] Infection via this route is common in rural areas with housing vulnerable to triatomine infestation or invasion, and animal host reservoirs living in close proximity to humans.[6] However, vector-borne transmission has declined markedly since 1990 due to extensive vector control programs throughout Latin America.[7] As vector-borne transmission declines, congenital transmission accounts for a larger proportion of overall new *Tc* infections. In addition, global migration into urban areas has shifted a substantial proportion of the disease burden into cities.[5]

During pregnancy, *Tc* parasitemia levels tend to rise around the end of the second trimester or during the third trimester when the maternal immune system is most modulated to prevent fetal rejection.[8,9] Furthermore, *Tc* has been shown to exhibit tissue tropism for placental tissues.[10] Cohort studies of congenital transmission of *Tc* suggest that women with less vector exposure may be more likely to transmit the parasite to their infant. Repeated exposure to *Tc* in endemic areas may prime women's immune systems to control parasitemia during pregnancy thus lowering the risk of vertical transmission.[11-15] Reported rates of cCD from 12 studies in six countries are highly variable, ranging from 2% to 14% in observed populations of seropositive mothers.[16,17] This variability may be due to parasite strain diversity, cumulative incidence of vector exposure in the maternal population, intensity of follow-up infant testing, and/or the sensitivity of the diagnostic testing algorithm employed. A complex interaction between parasite, placenta, and inflammatory and oxidative stress from repeat innate immune activation by increased *Tc* parasitemia is believed to weaken the placental barrier, allowing the parasite to cross into the fetal blood.[9,18] Whether and how environmental exposure stressors that elicit similar inflammatory immune reactions during pregnancy may cumulatively affect congenital transmission and neonatal development has not been characterized.[12,13] Little is known about the effect of exposure to air particulate matter (specifically particulate matter <2·5 μm diameter [$PM_{2\cdot5}$]), high temperatures, or the mixture of these exposures on risk of cCD.

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Environmental exposures during pregnancy, such as excessive $PM_{2\cdot5}$ and extreme temperatures, can have deleterious effects on birth outcomes. $PM_{2\cdot5}$ can cause placental degradation and adverse birth outcomes.[20-22] Studies indicate that a 1°C increase in global temperature may double the risk for both preterm birth and stillbirth.[23] Like mechanisms of harm from *Tc,* proposed mechanisms for associations of $PM_{2\cdot5}$ and temperature with adverse birth outcomes include inflammation and oxidative stress.[20-22] For example, $PM_{2\cdot5}$ can cross the placental barrier, resulting in placental weakening and maladaptation, with further deleterious effects on fetal development.[20] However, to date, the impact of the mixture of $PM_{2\cdot5}$ and heat exposures during pregnancy on congenital transmission of *Tc* has not been studied.

Bolivia has significant seasonal fluctuations in both wet bulb temperature (WBT) (10.6ºC-22.7ºC) and $PM_{2\cdot5}$ (2.3μg/m^3-93.7μg/m^3) (appendix pg.2-3). $PM_{2\cdot5}$ air pollution is largely driven by seasonal wildfires originating from agriculturally-related slash and burn activities.[19] To determine the effects of $PM_{2\cdot5}$ and high temperature exposure during pregnancy on cCD, we used Lagged-Weighted Quantile Sum (L-WQS) regression analysis to understand the joint effects of $PM_{2.5}$ and temperature throughout pregnancy and to identify critical windows of environmental exposure associated with cCD. Model inference systems using Weighted Quantile Sum (WQS) regression analysis have been used to analyze complex chemical mixtures present in environmental exposures and estimate their combined effects on birth outcomes.[25,26] Lagged-WQS allows for smoothed associations over time, similar to distributed lagged models, accounting for a mixture effect, with a WQS model run at each time point (appendix pg.4).[27] We evaluated the associations of monthly temperature exposure (WBT) and air pollution ($PM_{2\cdot5}$) with Tc transmission in a cohort of *Tc(+)* pregnancies in Santa Cruz and Camiri, Bolivia (2006-2014).[11,12,24] To our knowledge, this is the first study to assess the relationship between environmental exposures and congenital Chagas transmission. Additionally, this study represents the first time L-WQS has been used to quantify the potential effects of environmental exposure mixtures on the risk of congenital infection of an infectious disease and provides insight into the multiple influences underlying cCD transmission.

Methods

Setting

Maternal and infant data were derived from two cCD cohort studies conducted from 2006-2007 in Hospital Universitario Japones (HUJ), Santa Cruz de la Sierra and from 2010-2014 in HUJ and the Municipal Hospital of Camiri.[11,12,24] Both are regional referral hospitals with wide catchment areas. Santa Cruz is Bolivia's economic center with a population of ~1·7 million residents when the studies were conducted.[12] Camiri is a much smaller city with a population of 30,191 residents.[12] While there is generally no vector transmission of *Tc* in highly urbanized areas, both Santa Cruz and Camiri are destinations for migrants from rural areas with high rates of triatomine infestation and *Tc* transmission. In 2010, Bolivia had a estimated national *Tc* prevalence of 6% with prevalence of >50% in some rural areas.[28] In rural surveys in the hyperendemic Chaco ecozone, infection prevalences as high as 80% of adults and >60% of women of childbearing age were reported.[28-30]

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Ethical approvals and study designs

Protocols for cohort recruitments were approved by the institutional review boards of the Johns Hopkins Bloomberg School of Public Health, Hospital Universitario Japonés, Universidad Católica Boliviana, Universidad Peruana Cayetano Heredia, Asociación Benéfica Proyectos en Informatica, Salud, Medicina y Agricultura, and the Centers for Disease Control and Prevention. Each woman provided written informed consent for herself and infant. The secondary data analysis evaluating exposures with increased risk of cCD transmission was approved by the institutional review board of the Icahn School of Medicine at Mount Sinai.

Women were enrolled when they presented to give birth, completed a brief structured questionnaire and had blood screened for chronic Chagas disease by one or more rapid diagnostic tests based on IgG to *Tc*.[11,12,24] Rapid-test results were confirmed by at least two conventional IgG serological assays. Those whose rapid test results were positive or equivocal were asked to participate in a further study component that comprised screening and follow-up of their infants for up to one year. cCD was defined based on composite results of microscopy and PCR of cord blood and umbilical tissue and blood samples at 1-3 months of age, plus conventional IgG serology of infant serum at 9-12 months of age.

Environmental Exposures

European Space Agency (ESA) Copernicus climate data is available as a 0·75º (~80x80km) gridded product. Data for $PM_{2\cdot5}$ and WBT are available hourly from 1940 to present. These data were downloaded from the Copernicus Climate Change Service (C3S) Climate Data Store (CDS) for the entirety of Bolivia for 2006-2014 (appendix pg.5).[31,32] Hourly values were converted to daily mean estimates of $PM_{2\cdot5}$ and WBT for each grid cell. The daily mean estimates of the four cells intersecting the estimated hospital catchment zone (50 km radius or <1 hr drive) in Santa Cruz were calculated for both $PM_{2\cdot5}$ and WBT and converted to monthly exposures for the pregnancies in the Santa Cruz metropolitan area (Figure 1). Monthly pregnancy exposures were defined as the 30-day increments prior to delivery or Months Prior to Delivery (MPD) (appendix pg.6). Camiri's monthly exposure values were derived from a single cell (Figure 1). Camiri is estimated to have a smaller catchment zone due to the town's size and location in a valley surrounded by mountains and subtropical Chaco forests (Figure 1). MPD exposure estimates were then assigned to each pregnancy based on city and date of birth.

Univariate Analysis

We calculated descriptive statistics and conducted univariate analysis of dependent and explanatory variables as well as temporal environmental exposures. To assess if seasonality persisted through anchoring exposures to date of birth, a Kruskal-Wallis test was performed comparing median monthly pregnancy exposures over time. Median monthly exposures relative to birth were compared between positive and negative infants using a Wilcoxon rank-sum test. Additional Wilcoxon rank-sum tests were performed to compare monthly exposures between male and female infants. Finally, Kruskal-Wallis tests were used to compare monthly exposures across maternal age groups (five-year groups for ages 10-50).

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Lagged-Weighted Quantile Sum Regression Analysis

We conducted L-WQS regression mixtures analysis to evaluate a potential time-varying mixture effect of heat exposure (WBT) and air pollution ($PM_{2\cdot5}$) on association with the binominal outcome variable (+/- for *Tc*), adjusting for maternal age, infant sex, and city of birth. Maternal age was highly correlated (cor: 0.7) with number of previous pregnancies and both are known to correlate with cCD.[12] To avoid redundant covariates in the L-WQS analysis we used maternal age as an indicator for both number of previous pregnancies and maternal age. L-WQS was conducted with measures of mean monthly $PM_{2\cdot5}$ and WBT exposures ranked into deciles during the 12 MPD.[31] Separate L-WQS models were estimated constrained to either the positive or negative directions. All models were run without partitioning the data due to small sample sizes and with 100 bootstraps. Critical windows were identified as time periods where the 95% confidence intervals did not include zero. A sensitivity analysis with the same model parameters using perturbed environmental exposure data (not concurrent with individual MPD exposures) was conducted to evaluate the timing and impact of exposures as well as the effect of twins. Analyses were performed using the gamm4 version 0.2-6 and L-WQS version 0.5.0 packages in R version 4.1.1 (Vienna, Austria).[33]

Results

Cohort Characteristics

Among the 581 *Tc(+)* pregnancies, there were 593 infants born between 2006 and 2014 (Table 1). Of these 581 *Tc(+)* pregnancies, 46 infants were identified with cCD (7·90%; 15 male, 31 female) (Table 1, Figure 2a-2b). Stratified by city, Santa Cruz, had 353 *Tc(+)* pregnancies and 36 *Tc(+)* infants, a 10·2% cCD rate while Camiri had 228 *Tc(+)* pregnancies and 10 *Tc(+)* infants, a 4·4% cCD rate (Table 1).

Table 1. Cohort characteristics.

City	Mean Maternal Age (Range)	*Tc(+)* Pregnancies (n)	Infants from *Tc(+)* pregnancies (n)	*Tc(+)* Infants (n)	*Tc(+)* Male Infants (n)	*Tc(+)* Female Infants (n)	Years
Santa Cruz	30 (13 - 47)	353	362	36 (10·2%)	13	23	2006 - 2007, 2010 - 2014
Camiri	29 (15 - 43)	228	231	10 (4·4%)	2	8	2012 - 2014
Overall	30 (13 - 47)	581	593	46 (7·9%)	15	31	2006 - 2007, 2010 - 2014

Environmental Exposures & Univariate Analysis

The cohort exhibited a relatively even distribution of cCD across months and years (Figure 2a - b). The proportion of *Tc(+)* infants was significantly different by infant sex with 4·8% among males (15/314) and 11·1% among females (31/279) ($\chi^2(1)=7\cdot4$, $p=0\cdot01$). Median maternal age was 24 for *Tc(+)* infants and 26·8 for *Tc(-)* infants of infected mothers (Wilcoxon, $p=0\cdot02$).

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Median maternal age was 26·7 for male infants and 26·1 for females (Wilcoxon, p=0·6). $PM_{2\cdot5}$ and WBT demonstrate seasonality within Santa Cruz and Camiri, with $PM_{2\cdot5}$ concentrations higher from July to December, and WBT higher from November to March (appendix pg.2-4). However, Kruskal-Wallis tests comparing median monthly exposures over time to assess if seasonality persisted despite anchoring monthly exposures to birth date suggested there was no statistically significant difference in median seasonal exposures for $PM_{2\cdot5}$ ($\chi^2(11)=5\cdot0$, $p=0\cdot9$) and WBT ($\chi^2(11)=7\cdot1$, $p=0\cdot8$). Wilcoxon rank-sum testing comparing median monthly exposures between positive and negative infants indicated that $PM_{2\cdot5}$ was significantly elevated at -11 MPD for pregnancies with *Tc(+)* infants and again immediately preceding birth (-2 MPD), (all $p<0\cdot05$). WBT was significantly higher pre-conception and during the first month of the pregnancy in *Tc(+)* infants (-11 to -8 MPD), (all $p<0\cdot05$). Wilcoxon rank-sum tests comparing monthly exposures between male and female infants showed no significant differences for either exposure variable (all $p<0\cdot05$). Kruskal-Wallis tests comparing monthly exposures across maternal age showed no significant differences for either exposure variable (all $p<0\cdot05$).

Lagged-Weighted Quantile Sum Regression Analysis

L-WQS was conducted using WQS estimates from gWQS analyses of risk of *Tc(+)* conducted per time bin (here, per MPB) adjusted by covariates. With the binary outcome, the estimated beta coefficient is the difference in WQS values between those infants with and without *Tc.* The results showed an indication that the WQS index was higher in the *Tc(+)* group during pre-conception. During pre-conception, the weighted exposure index related to cCD begins at -11 until -9 MPD (Figure 3a). At -11 MPD, WBT and $PM_{2\cdot5}$ each contribute ~50% to the association while at -10 MPD until conception, WBT plays a larger role than $PM_{2\cdot5}$ (80% and 20%) (Figure 3b-c). The opposite is observed late in gestation. Gestationally, the positive trend begins in -4 MPD with a significant critical window [point where confidence intervals (CI) were above 0] between -2 MPD and birth (Figure 3a). WBT contributed to ~20% of the weighted association with cCD and $PM_{2\cdot5}$ ~80% in this critical window period at the end of pregnancy (Figure 3b-c).

Sensitivity analysis found L-WQS to show no association between randomly perturbed environmental exposure data and the risk of congenital transmission of *Tc* (Figure 3d) (appendix pg.8). This provides confidence that the association observed between $PM_{2\cdot5}$, WBT, and cCD is not random but rather represents a statistically significant association with cCD transmission. Sensitivity analysis conducting L-WQS on pregnancy rather than infant showed a similar model trend and weights as the analysis on infants but with reduced significance. Controlling for sex of the infant was important, given that in this cohort, the likelihood of congenital transmission was 2.3 times higher in female infants compared with male infants (appendix pg.8-9).

Discussion

Exposure to $PM_{2\cdot5}$ and WBT during the three months prior to conception, -9 MPD, and -3 to 0 MPD had significant associations with cCD. The immune responses to pre-conception exposure to the mixture of $PM_{2\cdot5}$ and WBT likely affect placental development while the immune responses to the mixture of $PM_{2\cdot5}$, WBT, and increased *Tc* parasitemia then weaken the placental barrier in late pregnancy. This combination of mechanical and inflammatory insults to the

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placenta may allow the parasite to more easily cross the marginal sinus and the trophoblastic layer of the intervillous space of the placenta and infect the fetus (Figure 4).

The relationships between environmental exposures and human health outcomes have complicated correlation patterns and may result in mixture effects. That is, individual exposures may be below a measurable effect level, but joint action of the components may produce significant effects. Ignoring joint action may lead to significant underestimation of risk. However, WQS allows examination of the exposures jointly to assess whether there is an increase in the risk of an adverse outcome such as cCD. The utilization of a weighted index to determine if the identified components significantly increase the adverse outcome uses a single degree of freedom test, which is the most powerful test of association. The method also accommodates collinearity in high-dimensional datasets by estimating parameters (weights used to construct the WQS index) across the varying conditions created in each bootstrapped sample. The novelty in our approach characterizes the time varying association of the mixture effect. Interestingly here, the relative importance of $PM_{2\cdot5}$ and WBT changed between pre-conception and late gestation.

In extreme instances pre-conception exposure to $PM_{2\cdot5}$ and heat can result in miscarriage, pre-eclampsia, or fetal development issues.[20-23] Exposure to $PM_{2.5}$ and heat may also predispose to vertical transmission of infectious diseases, as evidence suggests they also influence placental development and the intrauterine environment potentially making the placenta more vulnerable to invasion by *Tc* when parasitemia increases during the second or third trimester of pregnancy. The effects of pre-conception parental environmental exposure mixtures on the health of a pregnancy, fetus, and maternal-fetal interface structures are admittedly both understudied and difficult to study. However, chronic heat exposure can result in intrauterine inflammation that may affect the formation and integrity of the placenta.[22] $PM_{2\cdot5}$ exposure can similarly cause systemic chronic inflammation increasing the likelihood of, among others, diabetes, ovarian fibroids, and modulation of trophoblast growth and differentiation during placental formation.[34,35] Furthermore, $PM_{2\cdot5}$ can result in both ovarian dysfunction and male reproductive toxicity including changes in the sperm epigenome associated with placental dysfunction.[34,36-39] Combined, these pre-conception and early gestation effects may reduce placental integrity and further modulate the maternal immune system though chronic inflammatory pathways making it easier for *Tc* to cross the placenta.

Invasion of the placenta by *Tc* causes inflammatory and oxidative stressors in the placenta from repeat innate immune activation.[9,18] Paradoxically, this immune response aimed at controlling *Tc* can weaken the placental barriers such as the trophoblast layer allowing *Tc* to cross the marginal sinus and the trophoblastic layer of the intervillous space into the fetal capillaries and infect the fetus.[9,18] Similarly, circulating $PM_{2\cdot5}$ can cross the placental barrier, adversely alter placental gene expression and fetal growth regulation, and also directly weaken the placental trophoblastic layer.[20,40] In fact, in mice, placental exposure to $PM_{2\cdot5}$ has been demonstrated to induce intrauterine growth restriction and impaired placental trophoblast syncytialization through enhanced PGRN and mTOR signaling.[40,41] We hypothesize that exposure to high levels of $PM_{2\cdot5}$ during the third trimester of pregnancy increases maternal blood concentrations of $PM_{2\cdot5}$ leading to increased amounts of $PM_{2\cdot5}$ crossing the placental barrier. The subsequent immune reaction weakens the placental trophoblastic layer by impairing trophoblast syncytialization. The reduced

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connection between the trophoblast cells of this layer effectively makes the last structural barrier between the parasite on the maternal side of the placenta and the fetal venous network porous. Degradation of this layer would make it easier for the parasite to pass through and invade the fetal blood supply.

Additionally, exposure to high temperature events results in systemic inflammatory stress including in the uterine and placental structures.[22] Therefore exposure to higher temperatures during the third trimester at the same time as peak maternal immune system modification, increased parasitemia, and $PM_{2\cdot5}$ exposure likely serves to compound degradation of the placental structures and the effectiveness of the maternal immune system at controlling *Tc* and preventing invasion of the fetal venous network.

This study had several limitations given this is secondary data analysis. Gestational age was not always known or documented so we estimated date of conception based on date of birth to determine the timing of exposures. This may introduce error by over or underestimating some ages. Additionally, we noted a difference in the sex ratio of *Tc(+)* infants with significantly more female infants infected congenitally. This has not been seen in other cohorts and caution should be used in drawing conclusions for two reasons: our small overall sample size of *Tc(+)* infants and not having a value for the overall number of infants by sex born at the two hospitals during cohort recruitment. Additionally, cCD cohorts in Bolivia have a high rate of loss to follow up (LTFU) for visits following birth to 9 month months of age, which means some cCD cases were likely missed (estimated to be 25-33%; unpublished data) because qPCR and early serological tests are not 100% sensitive. Thus, we likely underestimate cCD incidence due to LTFU. Finally, our spatial resolution for exposure analysis was limited by only having city level locations for pregnancy periods and by the spatial resolution of the available measures of $PM_{2\cdot5}$. There were no ground-based measures of $PM_{2\cdot5}$ available in Bolivia at the time of these cohorts, and the Copernicus grid cells were at a 0·75º scale. Finer scale resolution would allow more personalized exposure profiles to each pregnancy for a more refined mixture analysis. However, using the WQS method we adjust our exposures to an index which allows us to compare high to high, high to low, and low to low exposures against each other, allowing us to account for the macro trends in exposure. Here we show this method will work at the macro-exposure level, but it may also work at the micro- or personal exposure level. More work needs to be done to evaluate the methods validity at the micro-level and potentially develop a more robust exposure risk model.

This work demonstrates that exposure to a mixture of elevated WBT and $PM_{2\cdot5}$ during the 90 days prior to conception, the first month of gestation, and during the third trimester each had associations with cCD. As rural to urban migration continues, cCD will constitute more of overall *Tc* transmission and planetary increases in $PM_{2\cdot5}$ and high-heat events may play a greater role in *Tc* transmission. For example, temperatures were the hottest on record in 2023, and Bolivia had the largest fires and highest $PM_{2\cdot5}$ levels ever recorded in 2025.[19,42-44] The association of gestational $PM_{2\cdot5}$ and temperature with cCD may serve as a model for other congenital infections and planning for future prevention studies as infections also spread to new regions around the world because of human migration and climate change. Furthermore, these results emphasize the importance of testing neonates and children born outside endemic countries to mothers from *Tc* endemic countries.[45] In urban areas, *Tc* transmission has shifted away from the vector; however, there is still a significant incidence of congenital transmission.

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Congenital transmission also occurs in countries with a substantial population of Latin American migrants where vector-borne *Tc* has been rare or absent, such as the United States, Spain, or Italy (Figure 1). Here we highlight the need for increased global awareness of Chagas disease, as well as the breadth of the effects of human driven climate change, including on infectious disease transmission, and the urgent need for action to combat it.

Contributors

Specifically, MJW conceived and designed the study. MJW, CB, NMB, NBD, RHG, SN, ROW obtained funding for the study. CB, RHG, and the CDWGPB conceived & designed the original cohort studies. CB, RHG, MJ, VRR collected and cleaned the original cohort data. MJW, AB, CB, SGG, RK and AP accessed, prepared, and validated the cohort data for application in this specific study. MJW, NBD, RK, AP accessed, prepared, and validated the environmental exposure data. SM and CG developed the WQS and L-WQS analysis. MJW, NBD, SM, and AP analyzed the data. MJW wrote the manuscript. All authors contributed to interpreting the results and revising the manuscript and had final responsibility for the decision to submit for publication.

Role of the funding source

The funders of this study had no role in the study design, data collection, data analysis, data interpretation, or writing of this report.

Declaration of interests

Dr. Bern receives royalties from Wolters Kluwer for Chagas disease topics in UpToDate. The remaining authors have declared that no competing interests exist.

Data sharing

The environmental exposure data used in this study may be accessed through the Copernicus Climate Change Service (C3S) Climate Data Store (CDS) at https://cds.climate.copernicus.eu/ (appendix pg.5). The congenital Chagas case cohort data cannot be made publicly available but may be requested as a Limited Data Set under a Data Use Agreement by contacting Dr. Natalie M. Bowman (natalie_bowman@med.unc.edu). R codes to replicate the analysis are available at https://github.com/mjwa88/congenitalChagas_2006_2014.

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Figures & Legends to:

Maternal $PM_{2.5}$ and temperature exposures pre-conception and during pregnancy are associated with increased risk of congenital Chagas disease transmission.

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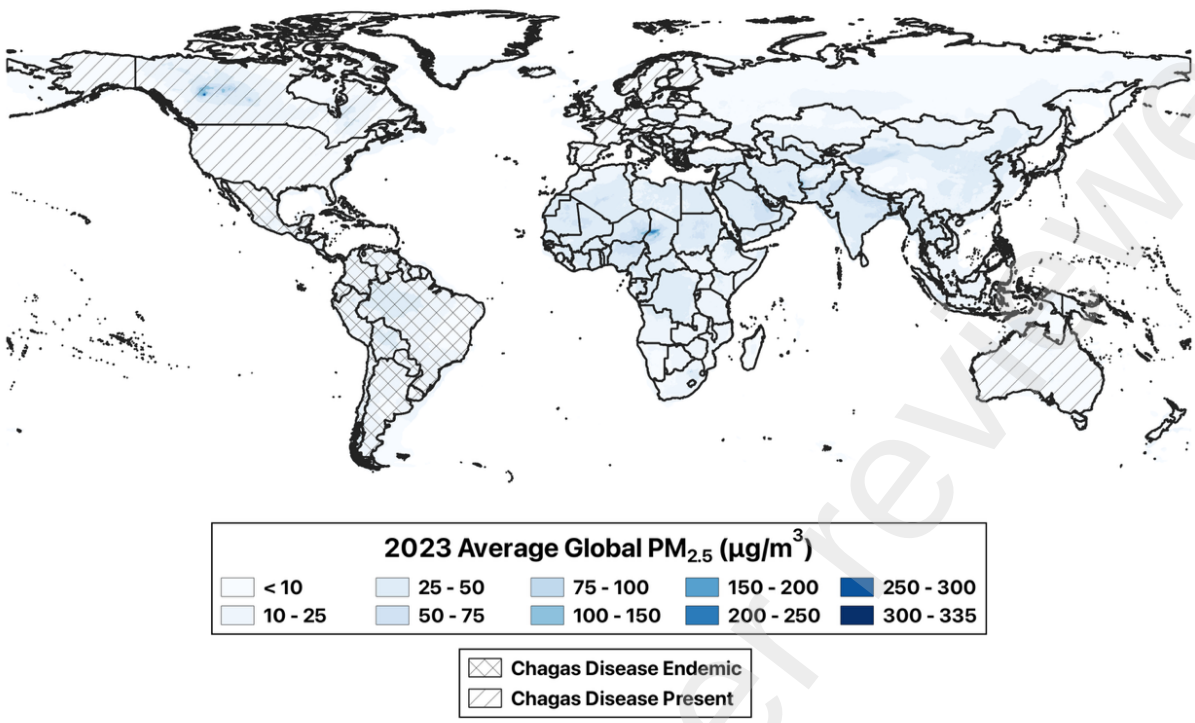

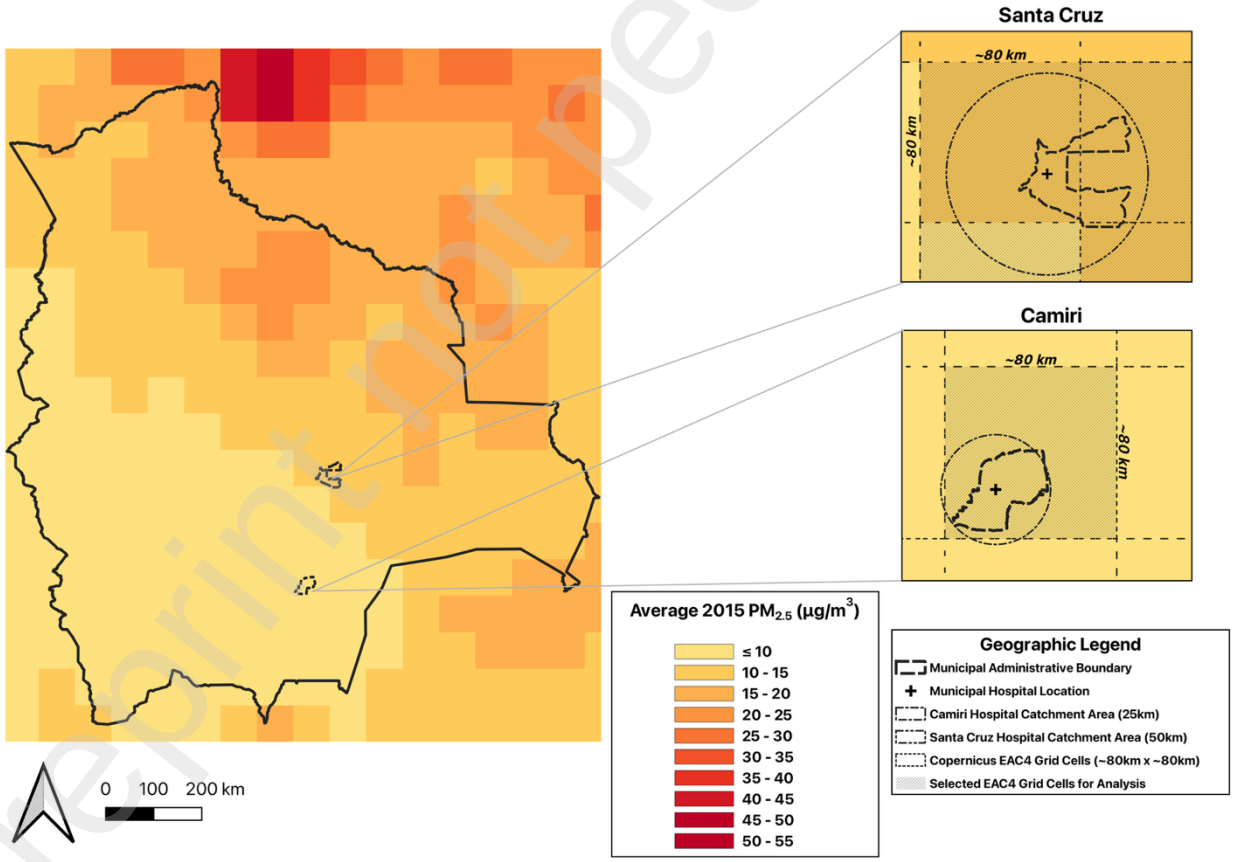

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Figure 1. Global distribution of Chagas infections, $PM_{2\cdot5}$, and cohort locations. **Top:** Global distribution of Chagas infections overlayed with 2023 average global $PM_{2\cdot5}$ levels. Cross Hashed: Endemic areas with vectoral transmission and Chagas infections. Hashed: Areas with Chagas infections but no known vectoral transmission. **Bottom:** Bolivia. Red Gradient: Mean monthly PM2·5 levels for 2015. Upper Inset: Santa Cruz municipality (dashed polygon), Hospital Universitario Japones (black cross), Copernicus grid cells averaged for exposure values (4 dotted squares), and 50 km catchment radius (dotted circle). Lower Inset: Camiri municipality (dashed polygon), Municipal Hospital of Camiri (black cross), Copernicus grid cell for exposure values (1 dotted square), and 50 km catchment radius (dotted circle).

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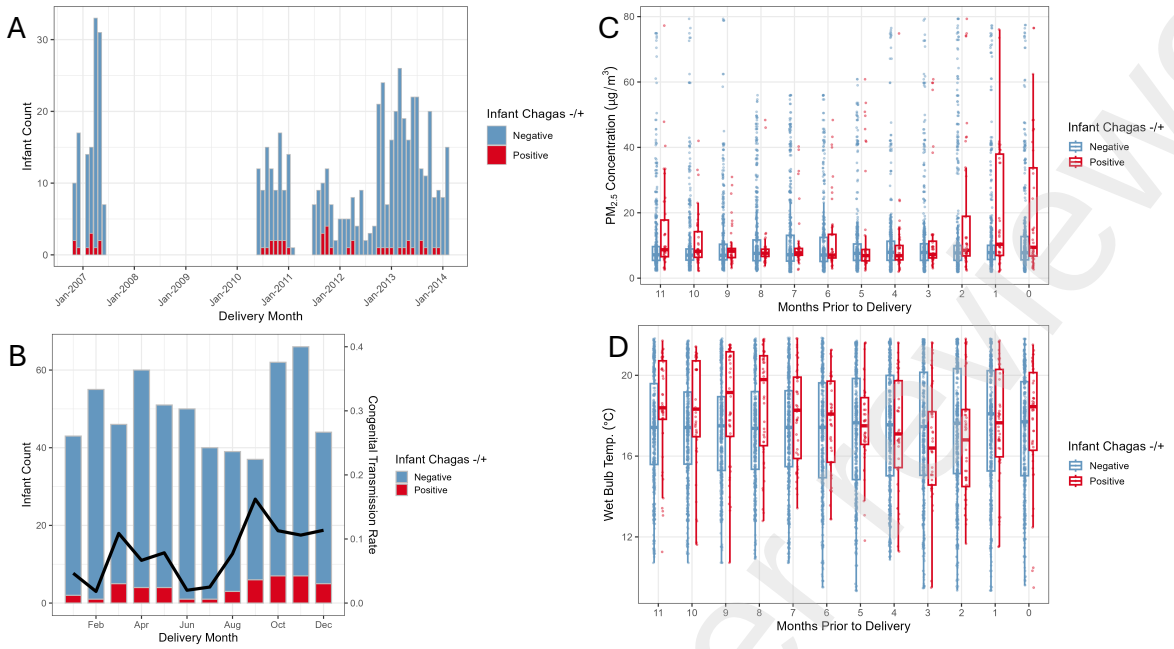

Figure 2. Panel A: Monthly distribution of *Tc(+)* (red) and *Tc(-)* (blue) infants over the course of the cohorts. **Panel B:** Total monthly distribution of *Tc(+)* (red) and *Tc(-)* infants over the course of the cohorts and overall monthly *Tc* infection rate (black line). **Panel C:** Distribution of pregnancy specific $PM_{2\cdot5}$ exposures by MPD of *Tc(+)* (red) and *Tc(-)* (blue) infants. **Panel D** Distribution of pregnancy specific WBT exposures by MPD of *Tc(+)* (red) and *Tc(-)* (blue) infants.

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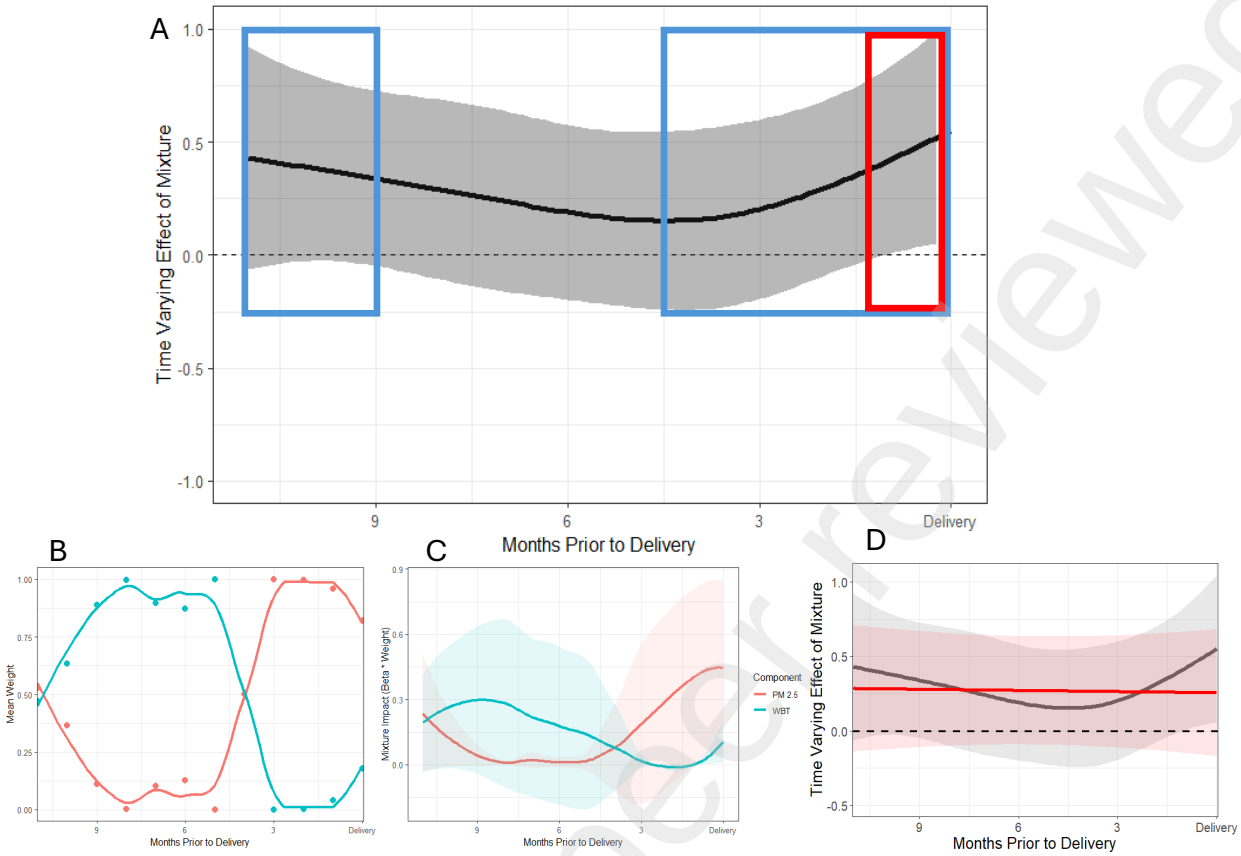

Figure 3. Panel A: Black line & grey CI: Significant association parameter with corresponding 95% CI. Blue boxes: Positive trends associated with increased cCD. Red box; significant critical window. **Panel B:** Estimated weights of each parameter at each time point in the L-WQS model (blue = WBT, red = $PM_{2\cdot5}$). **Panel C:** The weighted association (estimated weights multiplied by the association parameter at each time point) for both $PM_{2\cdot5}$ (red) and temperature (blue). **Panel D:** Black line & grey CI: Significant association parameter with corresponding 95% CI. Red line & red CI: Null model with perturbed environmental exposures.

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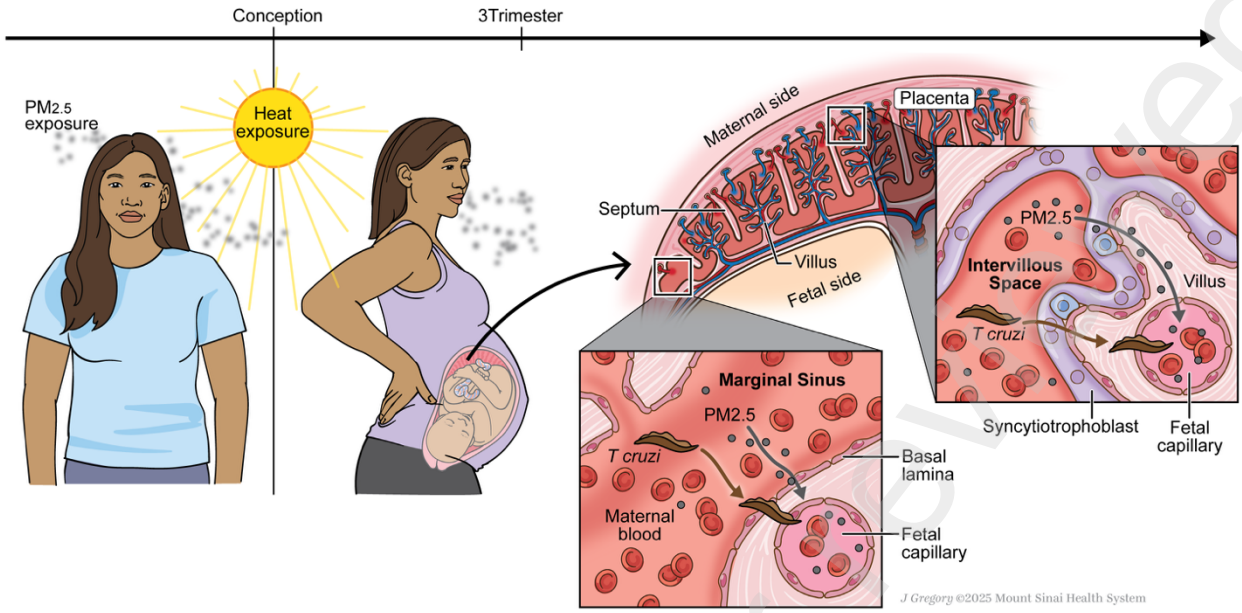

Figure 4. Timeline of exposures and illustration of parasite invasion of the fetal blood supply by crossing the basal lamina of the marginal sinus and trophoblast layer of the intervillous space.

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